

01. What is an ER Diagram?

An **ER (Entity-Relationship) diagram** is a visual tool used to model the structure of a database. It shows:

- **Entities** (like Student, Course),
- Their **attributes** (like name, roll, marks),
- And the **relationships** between them (like a Student "enrolls" in a Course).

It's widely used in the **design phase** of database development.

02. Different Types of Attributes (with examples)

1. **Simple Attribute:** Cannot be divided further.
Example: Age, Roll number.
 2. **Composite Attribute:** Can be divided into subparts.
Example: Name → First Name + Last Name.
 3. **Derived Attribute:** Value is derived from other attributes.
Example: Age (from DOB).
 4. **Multivalued Attribute:** Can have multiple values for one entity.
Example: Phone Numbers for a person.
 5. **Single-valued Attribute:** Only one value.
Example: Date of Birth.
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03. Total and Partial Participation

- **Total Participation:** Every entity instance must participate in the relationship.
Example: Every student must be enrolled in at least one course.
(Represented with a double line in ER diagram)
 - **Partial Participation:** Some instances may not participate.
Example: Some teachers may not supervise any project.
(Single line in ER diagram)
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04. Define Cardinality

Cardinality describes the number of instances involved in a relationship:

- **One-to-One (1:1):** One employee has one parking spot.
- **One-to-Many (1:N):** One teacher teaches many students.
- **Many-to-One (N:1):** Many students are supervised by one teacher.

- **Many-to-Many (M:N):** Many students can enroll in many courses.
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05. Comparison of Keys

- **Super Key:** A set of attributes that uniquely identify a row.
Example: (Roll, Name), (Roll), etc.
 - **Candidate Key:** Minimal Super Key — no unnecessary attributes.
Example: (Roll) is a candidate key.
 - **Primary Key:** Selected candidate key used for identifying records.
Only one per table.
 - **Foreign Key:** Refers to the primary key of another table to maintain relationships.
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06. Extended ER Diagram

An **Extended ER (EER) diagram** adds advanced modeling concepts:

- **Generalization:** Combining multiple entities into a higher-level entity.
Example: Car and Truck → Vehicle.
 - **Specialization:** Breaking an entity into sub-entities.
Example: Employee → Teacher, Admin.
 - **Aggregation:** Treating a relationship as an entity.
Example: "Project supervision" is treated as an entity in itself.
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07. Steps to Construct an ER Diagram

1. **Identify Entities:** What things do we want to store? (Student, Course)
2. **Identify Relationships:** How do entities interact? (Student *enrolls* in Course)
3. **Identify Attributes:** Properties of each entity (Name, Roll)
4. **Determine Keys:** Choose primary key for each entity.
5. **Draw Diagram:** Use symbols:
 - Rectangle = Entity
 - Oval = Attribute
 - Diamond = Relationship
 - Lines connect them